

	BIW	SOP #	SOP/TARA/SBU/GRINDING /01
		Implementation Date #	20.09.2025
		Revision #	00
		Last Revision #	15.09.2025
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SOP Owner #	STEEL BU – Tarapur Plant		

STANDARD OPERATING PROCEDURE FOR GRINDING OPERATION

1. **PURPOSE:** In order to avoid accident while operating the Grinding Machine Operation
2. **SCOPE:** These instructions are applicable to all Designated Areas.
3. **RESPONSIBILITY:** Production Supervisors, Engineer/ MED /EED/ Supervisor/HOD
4. **SAFETY:** Use the proper PPEs, Power Connection and Wheel Guard etc.
5. **SAFE OPERATING PROCEDURE:**
 - 5.1) Use the Personal Protective Equipment's Safety Helmet, Nose Mask, Ear Plug, Cotton Hand gloves, Safety Goggle/Faceshield, Body Apron and Safety Shoe.



- 5.2) Check the electrical plug board before power on the machine.
- 5.3) Switch off the "ON / OFF "button in the grinder when not in use and while turn on the electric connection.
- 5.4) Switch off the power supply while adjusting or changing new wheel.
- 5.5) Check the speed (RPM –Revolution Per Minute) of the machine and the speed of the wheel mentioned are matching before working with it.

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- 5.6) Clean and check the new wheel before mounting for any damage.
- 5.7) Make sure there is large washer on each side of the wheel and mounted properly.
- 5.8) Use only the appropriate key supplied with the machine to tight the wheel, ensure that the direction of tightening the flange nut opposite to the direction of rotation of the wheel and just tightened sufficiently.
- 5.9) While grinding hold the machine as the ground out hot particles hit directly on the floor, and make sure that can't reach any flammable material and other persons who work nearby.
- 6.0) If notice any abnormal sound or any other abnormalities don't use the machine, immediately inform to supervisor for rectification.
- 6.1) Don't remove the guard and back from the machine.
- 6.2) Use Fire Fighting Equipment's in case of emergency.
- 6.3) Disconnect the plug form the power source before making any adjustments, changing accessories, or storing the tool. Such preventive safety measures reduce the risk of starting the tool accidentally
- 6.4) At the time of work.
- Do not clamp portable grinders in a vise for grinding hand-held work.
 - Do not keep any materials close to the grinding wheel when it is not in use.
 - Do not carry a grinder by the power cord.
 - Do not pull on the power cord to disconnect it from the power supply.
 - Keep cords away from heat, oil and sharp objects or edges.
- 6.5) Store portable grinders on racks or hooks, not on the floor or ground.
- 6.6) Clean and service grinders according to the manufacturer's recommendations, including lubricating and changing accessories.
- 6.7) Keep a written maintenance record on portable grinders.
- 6.8) Verify the maximum RPM rating of the grinder does not exceed the RPM rating on each abrasive wheel.
- 6.9) When using a grinder that can switch between left-handed and right-handed operation, remember to move the blade guard when you move the handle.
- 7.0) Keep the power cord away from the grinding wheel and the material being ground.
- 7.1) Secure work with clamps or a vice to free both hands to operate the tool.
- 7.2) Tag out grinder if a deficiency is found – do not use grinder until the deficiency has been corrected.

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